Sliding, swinging and rolling

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Learning target: Choose the energy terms that match the type of motion.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 7.*
+240 J; the object gains gravitational potential energy.

02. Worked example: explain the model
Initial U 20; final rotational K 8 and translational K 12; all must sum consistently.

03. Read and interpret the model
Some energy is rotational kinetic energy, leaving less for center translation.

04. Apply the physics
15 J.

05. Explain your reasoning
Tension does zero work; gravity changes K.

06. Review: connect an earlier idea *Review: Unit 3, Day 13.*
Shift 24 m. Assumes a constant added transverse velocity and a straight-line comparison.

07. Exit check
The wheel spins, so it also has rotational kinetic energy.

08. STRETCH: test the model
Rolling stores part in rotation; sliding can put more into translation. Models and rotational properties matter.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 7.*
ΔU=-240 J; gravity does +240 J work.

02. Worked example: explain the model
Initial U 20; final rotational K 8 and translational K 12; all must sum consistently.

03. Read and interpret the model
Some energy is rotational kinetic energy, leaving less for center translation.

04. Apply the physics
mgh=0.75 mv²; v=√(4 gh/3)=√20=4.47 m/s. Point slider would√30=5.48 m/s.

05. Explain your reasoning
Tension does zero work; gravity changes K.

06. Review: connect an earlier idea *Review: Unit 3, Day 13.*
Shift multiplier 6/4≈1.5 because shift=(J/M) t. Assumes constant added transverse velocity.

07. Exit check
4.0 m/s.

08. STRETCH: test the model
mgh = ½mv² + ½mr²(v/r)² = mv²; v = √gh = √15 ≈ 3.87 m/s. The cylinder is faster at √20 ≈ 4.47 m/s because a smaller fraction of its energy is rotational.